	UNIVERSITY OF NUEVA CACERES **COURSE ASSESSMENT & EVALUATION PLAN** Form **Office of the Vice President for Academic Affairs**	**Doc. Control No.:** **UNC-FM-VPAA-02**

Name of Faculty:	NORTON, MONICA		**School Year**	2026-2027
Course:	BIT312L - Integrative Programming and Technologies (Revision 1)		**Semester**	1st

Course Outcome Statement		**Intended Learning Outcome**	**Assessment Tool**	**Performance Target**
1	Analyze integrative programming paradigms, language interoperability mechanisms, and structured data exchange standards.	Cite the value and relevance of the University's and the College's VMO as related to the course.	—	At least 90% of enrolled students with a rating of at least 60% of the total score
		Compare compiled, interpreted, and managed runtime environments and Foreign Function Interface (FFI) binding mechanisms.	• Objective-Type Quiz	At least 90% of enrolled students with a rating of at least 60% of the total score
		Parse, validate, and transform complex payloads across XML/XSD, JSON Schema, YAML, and Protocol Buffers.	• Polyglot Data Serialization Practicum	At least 90% of enrolled students with a rating of at least 60% of the total score
		Develop a cross-language data transformation and IPC bridge connecting Python, Node.js, and JVM/C# modules.	• Cross-Runtime Interoperability Utility	At least 90% of enrolled students with a rating of at least 60% of the total score

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2	Develop interoperable software components using REST, SOAP/WSDL, GraphQL, and polyglot scripting bridges.	Contrast SOAP/WSDL XML web services, RESTful resource APIs, and GraphQL query execution engines.	• Objective-Type Quiz	At least 90% of enrolled students with a rating of at least 60% of the total score
		Build a legacy SOAP-to-REST adapter service with automated WSDL parsing and JSON-Schema contract validation.	• SOAP-to-REST Protocol Adapter Lab	At least 90% of enrolled students with a rating of at least 60% of the total score
		Construct a unified polyglot BFF gateway aggregating REST, SOAP, and GraphQL upstream data sources.	• Midterm Polyglot API Gateway Project	At least 90% of enrolled students with a rating of at least 60% of the total score

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3	Implement secure third-party API integrations, webhooks, OAuth2 flows, and asynchronous queuing connectors.	Analyze OAuth 2.0 grant flows, HMAC webhook signature verification, and message queue worker topologies.	• Objective-Type Quiz	At least 90% of enrolled students with a rating of at least 60% of the total score
		Implement a resilient webhook receiver and OAuth2 client integrating payment, messaging, and cloud storage SDKs.	• Third-Party SDK & Webhook Integration Lab	At least 90% of enrolled students with a rating of at least 60% of the total score
		Engineer an asynchronous task-processing pipeline using Redis/Celery and RabbitMQ across polyglot worker nodes.	• Asynchronous Polyglot Worker Pipeline	At least 90% of enrolled students with a rating of at least 60% of the total score

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4	Build, containerize, and test a polyglot multi-tier application integrating heterogeneous services and cloud SDKs.	Evaluate container network bridges, reverse proxy routing, and end-to-end integration test harnesses.	• Objective-Type Quiz	At least 90% of enrolled students with a rating of at least 60% of the total score
		Configure multi-container Docker Compose environments with NGINX reverse proxies, TLS, and automated health checks.	• Multi-Container Reverse Proxy & CI Lab	At least 90% of enrolled students with a rating of at least 60% of the total score
		Present and defend a production-ready integrative software application uniting heterogeneous runtimes and external cloud APIs.	• Capstone Integrative Application Defense	At least 90% of enrolled students with a rating of at least 60% of the total score

Prepared by:	NORTON, MONICA		**Approved by:**	**DANILA, JUNAR**
Date Submitted:	November 11, 2026		**Date:**	November 11, 2026

Reminders: *This template should be accomplished for each course handled by the faculty.*
Course Outcomes and ILOs must be SMART; Each CO should be granularized into an introductory, enabling and demonstrative ILO; ILOs should NOT be teaching learning activities; Sample Performance target: At least 70% of students with 60% proficiency or score 12 out of 20.